openMPPT Hardware Design Calculations

Theoretical basis and calculations for component selection

1. Power Stage Parameters

Parameter	Symbol	Value	Notes
Switching Frequency	f_{sw}	100 kHz	Defined in `mppt.h` (`TIMER_PERIOD = 240`)
Maximum Input Voltage	$V_{in,max}$	80 V	Hardware limit
Maximum Output Current	$I_{out,max}$	20 A	Safety limit
Nominal System Voltage	V_{sys}	12V / 24V / 48V	Target batteries
Gate Drive Voltage	V_{drive}	10 V	Optimized for IRS21867 thermals (v1.3)

Planned change: 100kHz → 200kHz main switching frequency. Not yet implemented — `TIMER_PERIOD` in `mppt.h` (currently 240) and the table above still reflect 100kHz. `setup_cubemx_env_auto.py` already derives `TIMER_PERIOD` from `MPPT.ioc` and `main.c`'s PWM limits/dead-bands are dynamic, so the firmware side should follow from changing the timer config. See Section 9 for the full component re-check.

2. Gate Drive & Auxiliary Power Optimization (v1.3)

Primary Aux Supply (10V) - SCT2A25: Replaces the unstable 12V XL7005A with a robust 100V-rated step-down (SCT2A25).

- Output is tuned to **10V** instead of 12V.
- **Feedback Divider**: $V_{FB} = 1.2V$
  - $V_{OUT} = 1.2 \times \left(1 + \frac{R_{up}}{R_{down}}\right)$
  - Using E96 series: $R_{up} = 110\ k\Omega$, $R_{down} = 15\ k\Omega \rightarrow V_{OUT} = 1.2 \times \left(1 + \frac{110}{15}\right) = 10.0V$
- **Switching Frequency**: Fixed at 300kHz.
- **Inductor** (L_1): For 10V out, a standard **33µH or 47µH** inductor (e.g., 6x6mm shielded) is suitable.
- **UVLO Divider**: To set start at 12.5V: $R_{UVLO\_TOP} = 430\ k\Omega$, $R_{UVLO\_BOT} = 47\ k\Omega$ (Both are E24).
- **SW-node Freewheeling Diode**: D7, **SS510** (100V, 5A Schottky, C18199171) between the switch node and GND. (Previously documented here as "US1M" — never actually placed; verified against the schematic's real net connections.)

Secondary Aux Supply (3.3V Logic) - SY8120:

- Uses a "Cascaded Buck" architecture: 10V → Tiny Sync Buck (SY8120) → 3.3V.
- **Feedback Divider**: $V_{FB} = 0.6V$
  - $V_{OUT} = 0.6 \times \left(1 + \frac{R_{up}}{R_{down}}\right)$
  - Using E24 series: $R_{up} = 68\ k\Omega$, $R_{down} = 15\ k\Omega \rightarrow V_{OUT} = 0.6 \times \left(1 + \frac{68}{15}\right) = 3.32V \approx 3.3V$
- **Switching Frequency**: Fixed at 500kHz.
- **Inductor** (L): **4.7µH** (recommended for 3.3V out).

Aux power chain verified against datasheets. `SCT2A25STER`: input range 5.5-100V (20V headroom over the 80V bus, comfortable margin below the 12.5V min startup voltage), V_{FB} =1.2V±2% confirmed (validates the 110kΩ/15kΩ divider math above), 2A continuous/4A peak output vs. an estimated 150-500mA actual load (gate driver quiescent + switching current +

SY8120′s input-referred draw) — current isn't the sizing constraint, voltage range is. It's an **asynchronous** buck (only the high-side switch is integrated), which is why D7 (the freewheeling diode above) is required, not optional. Datasheet is marked "Preliminary Specification, Rev.0.8" by the manufacturer — worth knowing this is pre-release.

`SY8120B1ABC`: verified against the actual `SY8120B1` datasheet (matching LCSC C88474 exactly — the `SY8120E1ABC` datasheet used earlier was a same-family but different order-code stand-in, since superseded). V_{FB}=0.6V confirmed (validates the 68kΩ/15kΩ divider math above). Input range 4.5-18V vs. the 10V it actually sees (fed from SCT2A25′s output, not the 80V bus directly) — good margin both directions. Top FET current limit ($I_{LIM,TOP}$) is 2-3.6A, comfortably above L3′s 2A Isat — same category of fault-current gap as L2 had, but **not** upgraded: the 3.3V rail only realistically feeds the MCU/logic, nowhere near saturating L3 even with headroom for extras, unlike L2′s fan-load scenario. Decision, not an oversight.

Why D7 stays SS510, not consolidated to SS210 (unlike D1-D6, see `STANDARDS.md`): SCT2A25′s own 4A peak current limit exceeds SS210′s 2A continuous rating — under fault/inrush/heavy-transient conditions the diode could see close to that 4A, which would overstress SS210 but not SS510 (5A rated). D1-D6 avoided this problem because their worst-case stress is nanosecond-scale gate-charge pulses, not sustained fault current a diode has to survive thermally.

L2 changed for the same reason D7 stays SS510: the original `APS0650M470A` (47µH, Isat 2.62A, DCR 227mΩ) saturates well below SCT2A25′s 4A peak limit — fine for the 150-500mA baseline load, not fine if a heavier load (e.g. a fan) gets added to the 10V rail. Replaced with `GXDR1207-470MT` (FAUKU, C52196367, $0.207, 11735 in stock): same 47µH, Isat 4.0A (now matching the IC's own peak limit exactly instead of saturating below it), DCR 90mΩ (vs. 227mΩ — meaningfully lower conduction loss regardless of actual load). Package grew from 7.7×6.6mm to 12.5×12.5mm SMD.

3. Gate Driver / MOSFET Drive Synergy (IRS21867STRPBF, R36-R39, R9/R10/R17/R18)

Verified against the Infineon `IRS21867S` datasheet (full electrical table, not just the summary timing specs already checked in Section 9).

- **Supply range**: 6.8-20V vs. the 10V design point — comfortably mid-range, not a marginal operating point, so the propagation-delay specs (characterized at 15V) should hold closely at 10V too.
- **Driver output current**: I_{O+} (source) 5.34-6.66A, I_{O-} (sink) 4.90-6.10A typ/max — higher than the "4.0A typ" headline spec. With 10V drive through the 2.2Ω gate resistor (R36-R39), resistor-limited peak current is $10\frac{V}{2.2}\Omega \approx 4.55A$ — **below** the driver's 5.5-6A capability, meaning the resistor is deliberately the dominant current limiter (predictable dV/dt/EMI), not the driver's raw max rating.
- **Switching speed**: gate charge time ≈20-25ns (76nC max Q_g, current tapering from 4.55A toward 2.45A through the Miller plateau at $V_{plateau} \approx 4.6V$). Combined with the driver's 170-250ns propagation delay, total turn-on latency ≈200-275ns — about 4-5.5% of the 5000ns period at 200kHz, comfortable margin for dead-time budgeting.
- **100kΩ gate pulldown** (R9/R10/R17/R18): the driver's own UVLO drives HO/LO low when VCC/VBS is present-but-insufficient, but can't do anything with zero power (e.g. before the 10V rail comes up during power-on sequencing) — that's the specific gap this resistor covers. Negligible power draw during normal operation (100µA @ 10V); not intended for Miller-transient immunity during real switching (the driver's actively-low output handles that), just a defined off-state before the driver is powered. Sane, standard value for that job.

4. Transient Voltage Suppression (TVS) Strategy

Primary Bus Protection (80V):

- **Component**: **5.0SMDJ85CA** (Bidirectional, 5kW).
- **Rationale**: Standoff (V_{RWM}) of 85V ensures no leakage at 80V. 5kW rating handles massive inductive kickback and solar surges.
- **Placement**: Immediately adjacent to VIN/VOUT XT60 connectors.

Gate Drive Protection (10V):

- **Component**: **H12VH22U** (6kW Surge).
- **Rationale**: 12V standoff ensures invisibility at 10V rail. Protects sensitive IRS21867 drivers (25V max) from regulator failure.

Logic Protection (3.3V): removed (v1.3), D9 deleted.

- Previously **H3V3L06B**. Datasheet verification found it's an ESD-class part (DFN0603-2L, IEC 61000-4-2 rated, 80W / 8A pulse rating) rather than a power-rail transient absorber, and its own clamping-voltage table (3.5-6V at 1A, up to 10V at its rated 8A) exceeds the STM32′s 4.0V absolute max (confirmed against the STM32F072 datasheet, Table 21: Voltage characteristics) at any current beyond a trivial ESD-level event - it could not do the job its own rationale claimed.
- Checked purpose-built power-rail TVS alternatives before removing it: Nexperia PTVS3V3S1UR (400W) clamps at 8V, Littelfuse SMF3.3 (200-1200W) clamps at 6.8V. Neither clears 4.0V either - this is a physics limit of 3.3V-class TVS diodes (they need real separation from their own breakdown voltage to keep leakage low at rest), not a part-selection mistake. No drop-in replacement exists that actually satisfies the original "keeps the rail under 4.0V" rationale.
- Decision: remove D9 rather than keep a component that can't do its stated job. The 3.3V rail's fault protection now rests on the SY8120B1′s own regulation and the upstream Vin OV/UV limits already enforced in `system_config.h`, not an additional clamp on this rail.

5. Inductor Sizing (Main Power Stage)

Targeting a peak-to-peak ripple current (ΔI_L) of 20% of $I_{\text{out,max}}$ (**4.0 A**).

Buck Mode Worst-Case: (50% Duty Cycle, e.g., 80V in, 40V out)

$$L_{\text{buck}} = \frac{V_{\text{out}} \times (V_{\text{in}} - V_{\text{out}})}{\Delta I_L \times f_{\text{sw}} \times V_{\text{in}}}$$

At $f_{\text{sw}} = 100\text{kHz}$: $L_{\text{buck}} = \frac{40 \times 40}{4.0 \times 100000 \times 80} = *\, 50.0\mu\text{H}\, *$

v1.3 Component Selection: Target inductor size: **33 µH to 47 µH**.

- **CRITICAL**: Saturation Current (I_{sat}) **MUST be > 25A**. Using a lower I_{sat} (like 11A) will cause magnetic collapse and MOSFET destruction.

6. Bulk Capacitor Selection

Targeting an output voltage ripple (ΔV_{out}) of **< 100mV**.

$$\text{ESR}_{\text{max}} = \frac{\Delta V_{\text{out}}}{\Delta I_L} = \frac{0.1\text{V}}{4.0\text{A}} = *\, 25m\Omega\, *$$

v1.3 Capacitor Strategy:

- Use **4x 330µF 100V Low-ESR Electrolytic** (Ymin LKML2502A331MF, C443153) in parallel, per bulk bank (input and output each get their own 4-cap bank, C15-C18 / C20-C23 in the schematic).
- **Verified against the Ymin LKM series datasheet** (not assumed): the part number itself decodes to series LKM + diameter code L (12.5mm) + height code 250 (25mm) + voltage code 2A (100V) + capacitance code 331 (330µF) + M (±20%) - exactly LKML2502A331MF. The series' standard-item table confirms the 12.5×25mm/100V/330µF row directly: **ESR = 47mΩ max** and

ripple current = 2.14A rms, both specified at 100kHz/25°C - the exact figures this section already assumed.

- **Total Bank ESR**: $\frac{47m\Omega}{4} = *11.75m\Omega*$ (Passes the 25mΩ limit with 53% margin).
- **Total Ripple Capacity**: 2.14 A × 4 = ∗ 8.56 A ∗ per bank. The actual RMS ripple current a bulk cap bank sees is $\approx \frac{\Delta I_L}{2\sqrt{3}} \approx 1.15$ A - the 8.56A rating is headroom against that, not a comparison to the 20A DC bus current.
- **Voltage margin**: 100V rating on an 80V max bus is a 20% derating margin - inside the range electrolytics are normally run at continuously, though tighter than the 25-30% some DC-link guidance recommends.
- The series' own ripple-current frequency correction table (0.40, 0.50, 0.80, 0.90, **1.00** at 50Hz, 120Hz, 1kHz, 10-50kHz, 100kHz respectively) confirms the rated ripple current is already anchored at its ceiling by 100kHz, with no published derating beyond that - supporting the "frequency-independent" call in the Section 9 re-check below.
- **Cost pass (checked, not adopted)**: cheaper and SMD alternatives were surveyed against this same spec. Nothing beats LKML2502A331MF on cost, ESR, ripple, and life simultaneously - every cheaper part found trades away life rating (10000h to 5000h) to hit its price point. The one genuine free improvement: **3 caps per bank would already suffice** - $\frac{47m\Omega}{3} \approx 15.7m\Omega$ still clears the 25mΩ limit with 37% margin, and ripple capacity (2.14 A × 3 ≈ 6.4 A) stays far above the 1.15A actual RMS need. Staying at **4x** for the extra ESR/ ripple headroom and to keep it symmetric with the input/output bank layout - not a requirement, a margin choice.

7. Voltage Divider & ADC Scaling

Voltage Sensing LPF (v1.3 Optimized):

- $R_{\text{top}} = 200$ kΩ, $R_{\text{bottom}} = 4.7$ kΩ.
- $R_{\text{th}} \approx 4.59$ kΩ
- $C_{\text{filter}} = 10$ nF
- $f_c = \frac{1}{2\pi \times 4590 \times 10 \times 10^{-9}} \approx 3.47$ kHz
- *Evaluation*: Balanced for high-speed tracking and noise rejection.

8. Current Sense (CC6937S8-3FB020, U2/U3)

Verified against the CrossChip `CC6937` datasheet (not a name-convention guess).

- **Type**: Isolated Hall Effect, 3750V_{RMS} isolation (pins 1-4 to 5-8) — 47x margin over the 80V bus.
- **Sensitivity**: **66mV/A** (confirmed from the ordering table for the −3FB020 variant).
- **Supply**: VCC 3.0-3.3V nominal — the "3" in "3FB" denotes the 3.3V-supply series, matching the system's logic rail exactly.
- **Range**: ±20A. Zero-current output $V_{\text{OUTQ}} \approx 1.65$V (VCC/2, confirmed), full scale swings to ≈ 0.33-2.97V — centered and within a 3.3V ADC's linear range.
- **Response time**: 1.5µs (10-90% step) — well under one 200kHz switching period (5µs), adequate for fast overcurrent/backflow trip.
- **Bandwidth**: 230kHz −3dB, 150x margin over the 1.5kHz control loop sample rate. Margin over switching frequency shrinks from 2.3x (100kHz) to 1.15x (200kHz) at full power — not a concern for this design (average sensing + fast-trip response time, not raw bandwidth, govern both use cases here), but worth knowing if the sensor's role ever changes to reconstructing switching-ripple waveforms directly.
- **Insertion loss**: 0.5mΩ internal conductor resistance — negligible, and the reason this replaced the INA240+shunt approach from v1.1 (per `ROADMAP_V1.3.md`).
- **Decoupling**: **1nF** VREF-to-GND (matches the datasheet's own `CREF=1nF` test condition; confirmed in the schematic as C10/C26) and a separate **100nF (0.1µF)** VCC bypass (confirmed as C12 near U2). Also 1nF at VOUT per the datasheet's `COUT=1nF` condition.

9. 200kHz Migration: Component Re-Check

This section re-verifies the parts already selected for 100kHz operation against a doubled switching frequency, using values pulled directly from the vendored datasheets (not assumed). Nothing here has been implemented in firmware or hardware yet — this is the pre-check called for by the `STANDARDS.md` "Part selection & calculations" mandate before the `ROADMAP_V1.3.md` Phase 4 checklist item is closed.

9.1. Main Switching MOSFETs (Q1-Q4, BSC030N08NS5, C501507)

Source: Infineon `BSC030N08NS5` datasheet, Rev. 2.3, 2019-10-31.

Parameter	Value	Condition
$R_{\mathrm{DS(on)}}$	2.6 mΩ typ / 3.0 mΩ max	V_{GS}=10V, I_D=50A
Q_g (total gate charge)	61 nC typ / 76 nC max	V_{DD}=40V, I_D=50A, V_{GS}=0→10V
t_r (turn-on rise)	12 ns typ	V_{DD}=40V, V_{GS}=10V, I_D=50A, $R_{\mathrm{G,ext}}$ =3Ω
t_f (turn-off fall)	13 ns typ	same
C_{oss}	700 pF typ / 910 pF max	V_{DS}=40V, f=1MHz
$R_{\mathrm{th\ JA}}$ (PCB only, no heatsink)	50 K/W	6 cm² copper, 1-layer, natural convection
$T_{\mathrm{j,max}}$	150 °C	—

Conduction loss (frequency-independent):

$$P_{\mathrm{cond}} = I_{\mathrm{rms}}^2 \times R_{\mathrm{DS(on),max}} = 20^2 \times 0.0030 \approx * 1.2\ \mathrm{W} *$$

Switching loss (linear-approximation model, worst case $V_{\mathrm{DS}} = 80\mathrm{V}$, $I_D = 20\mathrm{A}$):

$$P_{\mathrm{sw}} = \frac{1}{2} \times V_{\mathrm{DS}} \times I_D \times (t_r + t_f) \times f_{\mathrm{sw}}$$

	100 kHz (current)	200 kHz (proposed)
P_{sw} per hard-switched MOSFET	2.0 W	4.0 W
P_{cond} per MOSFET	1.2 W	1.2 W (unchanged)
Total est. loss	**≈3.2 W**	**≈5.2 W**

Thermal flag. With PCB-only cooling ($R_{\mathrm{th\ JA}}$=50 K/W, no heatsink/thermal vias), the datasheet's own max package dissipation at T_A=25°C is $P_{\mathrm{tot}} = \frac{150-25}{50} = * 2.5\ \mathrm{W} *$ — already below the ≈3.2W estimate at **100kHz**, let alone ≈5.2W at 200kHz. This assumes worst-case V_{DS}/ I_D coincidence and a simplified switching-loss model (real loss depends on which of Q1-Q4 is hard-switched vs. synchronous-rectifying in the actual buck-boost topology, and gate resistance/ drive strength affect t_r/t_f in-circuit). This is exactly what `ROADMAP_V1.3.md` Phase 3′s open item ("20A Thermal Design: widen high-power traces, implement massive thermal via grid") already anticipates — that item should land **before** or **alongside** the 200kHz switch, not after. The bottom-side $R_{\mathrm{th\ JC}}$ path (0.5-0.9 K/W typ/max) is dramatically better than the PCB-only path and is what a proper thermal via array would unlock.

Gate drive loss (in the driver/gate resistor, not the MOSFET junction — informational):

$$P_{\mathrm{gate}} = Q_g \times V_{\mathrm{GS}} \times f_{\mathrm{sw}} = 76 \times 10^{-9} \times 10 \times f_{\mathrm{sw}}$$

At 100kHz: 76mW/MOSFET. At 200kHz: 152mW/MOSFET. Not a concern for the driver IC's thermal budget either way.

9.2. Gate Driver (IRS21867STRPBF, C52290)

Source: Infineon `IRS21867S` datasheet v01.00.

Parameter	Value	Condition
t_{on} / t_{off} propagation delay	170 ns typ / 250 ns max	$V_{\text{CC}} = V_{\text{BS}}$=15V, C_L=1000pF
MT (delay matching, $\lvert t_{\text{on}} - t_{\text{off}} \rvert$)	35 ns max	—
t_r / t_f (driver output)	22/18 ns typ, 38/30 ns max	—
Output source/sink current	4.0 A typ	—

At 200kHz the switching period is 5.0µs vs. 10µs at 100kHz. The 170-250ns propagation delay grows from 2.5% to 5% of the period — worth including explicitly in dead-time budgeting, but well within normal margins; **not** a hard blocker on 200kHz operation. Output drive current (4.0A) is unchanged and still comfortably swings Q_g=76nC in the tens-of-ns range required. No datasheet-stated maximum switching frequency applies here.

9.3. Main Inductor (L4)

Re-deriving Section 5′s formula at 200kHz:

$$L_{\text{buck}}(200\text{kHz}) = \frac{40 \times 40}{4.0 \times 200000 \times 80} = * \, 25.0\mu\text{H} \, *$$

This halves the 100kHz target (50µH → 25µH). The originally-placed part, `FC-SE2822-150M` (C46553544, verified via datasheet, not the name-convention guess this section originally used): **15µH ±20%, Isat 30.5A, rated 30A, DCR 3.5mΩ, 28x23mm THT, $2.30 / 407 in stock**.

	100kHz (current firmware)	200kHz (planned)
Ripple current ΔI_L	13.3A	6.67A
Peak current / I_{sat} margin	26.7A / 14%	23.3A / 31%
Output ripple ΔV_{out} (bank ESR 11.75mΩ)	**157mV — 57% over the 100mV target**	78mV ✓
Conduction loss ($I_{\text{out}}^2 \times$ DCR)	1.40W	1.40W (freq-independent)

L4 only meets the ripple spec at 200kHz — a real, accepted dependency, not a bug. It was deliberately sized on the assumption that the 200kHz switch happens, specifically to avoid the "humongous" 50µH part the 100kHz target would otherwise require. This is not something to fix by re-checking values; it's a documented design decision.

Candidate replacements evaluated — two rounds. First pass (22-30µH / Isat≥30A, filtered to in-stock — 19 of 21 candidates found were out of stock and excluded) optimized purely for margin and found `C37634008` (RSEQ32-220M, Ruishen): 22µH, Isat 90A, DCR 0.9mΩ, $6.26/26 in stock — 304% Isat margin and 0.36W conduction loss at 200kHz, a huge improvement. **Rejected after a second look**: this compared candidates against each other without ever checking the original part's own price/stock ($2.30, 407 in stock) — a 2.7x price and 15x stock gap that makes "maximum margin" the wrong optimization target for a prototype build.

Second pass widened the inductance range (15-33µH) and weighed cost/stock properly:

FC-SE2822-150M (orig.)	C54830921	**HCZH-3218-150-M (chosen)**	C37634008
Manufacturer FANGCHENG	Coilank	YOUCHI	Ruishen

	FC-SE2822-150M (orig.)	C54830921	**HCZH-3218-150-M (chosen)**	C37634008
Stock / Price	407 / $2.30	69 / $2.87	88 / $3.86	26 / $6.26
L / Isat / DCR	15µH / 30.5A / 3.5mΩ	15µH / 44A / 3.0mΩ	15µH / 30A / 2.2mΩ	22µH / 90A / 0.9mΩ
Package	THT 28x23mm	THT 33.5x28mm	THT 33.5x28mm	THT 33x31.5mm
200kHz ripple / Isat margin	78mV / 31%	78mV / 89%	78mV / 29%	53.4mV / 304%
200kHz conduction loss	1.40W	1.20W	**0.88W**	0.36W

`C54830921` is the best pure value play (+$0.57 for 89% Isat margin, same ripple). `HCZH-3218-150-M` was chosen instead for its lower DCR (0.88W conduction loss, a real efficiency win) at a still-modest $1.56 premium over the original part — a deliberate prototype-stage trade-off (moderate cost increase for a meaningful efficiency gain, not chasing the maximum-margin `C37634008` pick, which cost 2.7x more for headroom this design doesn't need).

Chosen: `HCZH-3218-150-M` (YOUCHI, `C53699952`). Applied to `buck_boost.kicad_sch`; PCB footprint not yet updated — see `STANDARDS.md`'s L4 note for status.

9.4. Summary

Item	**Verdict at 200kHz**
MOSFETs (BSC030N08NS5)	Electrically fine (Vds/Rds(on)/Qg headroom unchanged), but switching loss ≈doubles — thermal design (ROADMAP Phase 3) must land first
Gate driver (IRS21867STRPBF)	No concern — propagation delay and drive current have ample margin
Main inductor (L4)	Replacement chosen: `HCZH-3218-150-M` (C53699952) — clears ripple target at 200kHz, DCR cut from 3.5mΩ to 2.2mΩ (0.88W conduction loss), for a modest $1.56 premium over the original part
Bulk caps / TVS / voltage sensing	Frequency-independent, no re-check needed